ARAVIND NAIDU YALLA

AI SOFTWARE ENGINEER • PRODUCTION AI SYSTEMS

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PROFESSIONAL PROFILE

AI Software Engineer with 1 year of hands-on experience building and deploying AI-powered applications and backend systems. Experienced in Python, FastAPI, LLMs, RAG, embeddings, semantic search, Qdrant, Ollama, Hugging Face and PostgreSQL. At People Media Factory, worked across the development lifecycle—from understanding requirements and designing solutions to deployment, debugging and maintenance. Focused on turning real-world business requirements into practical software and AI solutions.

TECHNICAL SKILLS

AI & Generative AI:	Python, LLMs, Generative AI, RAG, Prompt Engineering, Embeddings, Semantic Search, Hugging Face, Ollama
Backend:	FastAPI, REST APIs, Flask, Pydantic, SQLAlchemy, Object-Oriented Programming (OOP)
Databases:	PostgreSQL, Qdrant (Vector Database), SQL
Engineering & Ops:	Git, GitHub, Debugging, Troubleshooting, API Design, IIS, NSSM
Learning / Exploring:	Agentic AI, LangChain, LangGraph, Tool Calling

PROFESSIONAL EXPERIENCE

People Media Factory — AI Developer / AI Software Engineer — Hyderabad, India

May 2026 – Present • Full-Time | October 2025 – April 2026 • Intern — Promoted: Intern → Full-Time

- Built and deployed AI-powered applications using Python, FastAPI, LLMs and RAG.
- Developed backend services and REST APIs for production applications.
- Built document intelligence workflows involving OCR, embeddings, semantic search and vector databases.
- Integrated Hugging Face models and locally hosted LLMs using Ollama.
- Worked with PostgreSQL for application data and query optimization.
- Diagnosed production issues and optimized backend/database performance.
- Deployed and maintained applications using IIS and NSSM.
- Worked with Git/GitHub and collaborated with cross-functional teams.

KEY PROJECTS

Enterprise RAG Chatbot — Document Intelligence — Python, FastAPI, RAG, Qdrant, Ollama, Hugging Face, OCR, Embeddings

- Built an end-to-end document question-answering system using Retrieval-Augmented Generation (RAG).
- Engineered workflow: Documents → OCR/Extraction → Text Processing → Chunking → Embeddings → Qdrant → Semantic Search → Relevant Context → LLM → Answer.
- Implemented multi-document ingestion, dense embeddings, Qdrant vector storage, and grounded question answering.

VFX Production Tracking System — Python, FastAPI, PostgreSQL, REST APIs, IIS, NSSM, Git

- Built a production tracking platform for managing VFX artists, tasks, statuses, working activity, and timesheets.
- Designed role-based access control, supervisor approval workflows, and optimized PostgreSQL schema.
- Deployed and managed production services on Windows Server using IIS and NSSM.

AI Music Studio — Python, FastAPI, Generative AI, LLMs, Audio AI

- Worked on AI-powered music and audio generation workflows involving text prompts, LLM integrations and backend APIs.
- Orchestrated asynchronous FastAPI endpoints to handle compute-intensive audio generation models.

EDUCATION & CERTIFICATIONS

B.Tech — Computer Science (Data Science) • CGPA: 7.5 / 10 — Dec 2021 – Apr 2025

Kakinada Institute of Engineering and Technology • Kakinada, Andhra Pradesh
Affiliated to Jawaharlal Nehru Technological University Kakinada (JNTUK)

- **Certifications:** Python for Data Science (Infosys Springboard) • Salesforce Developer (SmartInternz)